

Transformation Forms:

Cubic: $f(x) = a(b(x-h))^3 + k$

Cube Root: $f(x) = a\sqrt[3]{b(x-h)} + k = a(b(x-h))^{1/3} + k$

Parameter	Transformation	Effect
a	Vertical stretch/compress & reflection	$ a > 1$: steeper $ a < 1$: flatter $a < 0$: reflect over x -axis
b	Horizontal stretch/compress	$ b > 1$: compress by $\frac{1}{b}$ $ b < 1$: stretch by $\frac{1}{b}$
h	Horizontal shift	$h > 0$: right $h < 0$: left (<i>opposite of sign in equation</i>)
k	Vertical shift	$k > 0$: up $k < 0$: down

Why does b work “backwards”?

Think about it: In $f(x) = (2x)^3$, when $x = 1$, you're computing $f(1) = (2 \cdot 1)^3 = 2^3 = 8$.

You get the same y -value that the parent has at $x = 2$, but you reach it at $x = 1$ (half as far from the origin).

The b multiplies the input, so the graph reaches each y -value b times faster $\rightarrow$ horizontal *compression*.

If $b = \dots$	Then the graph is...	Example
$b = 2$	Horizontally compressed by factor of $\frac{1}{2}$ (half as wide)	$(2x)^3$
$b = 3$	Horizontally compressed by factor of $\frac{1}{3}$ (third as wide)	$(3x)^3$
$b = \frac{1}{2}$	Horizontally stretched by factor of 2 (twice as wide)	$(\frac{1}{2}x)^3$

Parent Function Anchor Points:

Cubic $y = x^3$	Cube Root $y = \sqrt[3]{x} = x^{1/3}$
$(-2, -8)$	$(-8, -2)$
$(-1, -1)$	$(-1, -1)$
$(0, 0)$	$(0, 0)$
$(1, 1)$	$(1, 1)$
$(2, 8)$	$(8, 2)$

Point Transformation Rule:

$$(x, y) \rightarrow \left(\frac{x}{b} + h, ay + k\right)$$

Horizontal: divide x by b , then add h

Vertical: multiply y by a , then add k

Center Point: Always at (h, k)

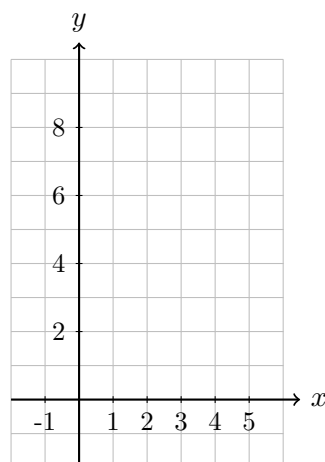
Example 1: $f(x) = 2\sqrt[3]{3(x-1)} + 4 = 2(3(x-1))^{1/3} + 4$

$a = \underline{\hspace{1cm}}$ $b = \underline{\hspace{1cm}}$ $h = \underline{\hspace{1cm}}$ $k = \underline{\hspace{1cm}}$

Center: $(\quad, \quad)$

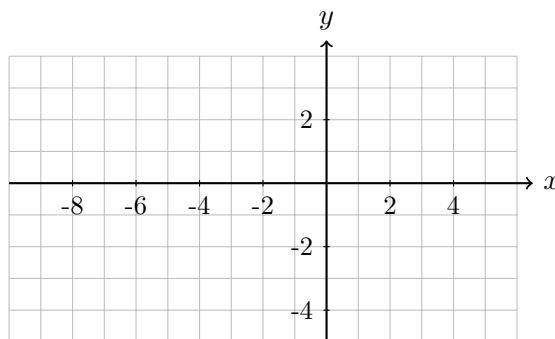
Parent	New
$(-8, -2)$	$(\quad, \quad)$
$(-1, -1)$	$(\quad, \quad)$
$(0, 0)$	$(\quad, \quad)$
$(1, 1)$	$(\quad, \quad)$
$(8, 2)$	$(\quad, \quad)$

y -int: _____



Example 2: $f(x) = -\frac{1}{2} \left(\frac{1}{2}(x+3) \right)^3 - 1$ $a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}} \quad h = \underline{\hspace{2cm}} \quad k = \underline{\hspace{2cm}}$ Center: ($\hspace{1cm}$, $\hspace{1cm}$)

Parent	New
$(-2, -8)$	($\hspace{1cm}$, $\hspace{1cm}$)
$(-1, -1)$	($\hspace{1cm}$, $\hspace{1cm}$)
$(0, 0)$	($\hspace{1cm}$, $\hspace{1cm}$)
$(1, 1)$	($\hspace{1cm}$, $\hspace{1cm}$)
$(2, 8)$	($\hspace{1cm}$, $\hspace{1cm}$)

 y -int: _____**Example 3: Write the equation from the description.**

A cubic function that is vertically stretched by 3, horizontally compressed by $\frac{1}{2}$, reflected over the x -axis, shifted right 2, and shifted down 4.

Identify each parameter: $a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}} \quad h = \underline{\hspace{2cm}} \quad k = \underline{\hspace{2cm}}$ $f(x) = \underline{\hspace{10cm}}$ **Example 4: Write the equation from the description.**

A cube root function that is vertically compressed by $\frac{1}{2}$, horizontally stretched by 4, shifted left 5, and shifted up 2.

Identify each parameter: $a = \underline{\hspace{2cm}} \quad b = \underline{\hspace{2cm}} \quad h = \underline{\hspace{2cm}} \quad k = \underline{\hspace{2cm}}$ $f(x) = \underline{\hspace{10cm}}$ **Vertical vs. Horizontal Transformations**

	Vertical (affects y)	Horizontal (affects x)
Parameters	a and k	b and h
Behavior	Does what you expect	<i>Opposite</i> of what you expect
Stretch/Compress	$ a > 1$ stretches	$ b > 1$ compresses

Memory trick: Horizontal transformations are “inside” the function, so they work backwards!**Rational Exponent Reminder:** $\sqrt[3]{x} = x^{1/3}$ and $\sqrt[3]{x^2} = x^{2/3}$ Example: $\sqrt[3]{8} = 8^{1/3} = (2^3)^{1/3} = 2^{3 \cdot \frac{1}{3}} = 2^1 = 2$